

**Memo 1: Background & Overall Mass and Energy Balances
Covering Chapters 1-3****Due Date: 02/03/2026****OVERVIEW**

This memo establishes the **foundation** for your entire reactor design project. You will conduct a literature review to understand your assigned petrochemical system and perform preliminary stoichiometric and energy balance calculations.

Memo 1 has two main sections:

SECTION 1: Literature Search and Process Background

SECTION 2: Preliminary Calculations

Grading:

- Literature review: 40 points
- Hand calculations: 30 points
- **Python exercise:** 20 points
- **Documentation & presentation:** 10 points
- **Total: 100 points**

Literature Search and Background**1.1 Process Background:**

This section forms the basis for the introduction of your final report. Initially, use the general references cited (Kirk-Othmer, McKetta, etc.) plus at least one recent patent on your technology.

Required Content:

- 1. Product Description & Uses**
 - Major industrial applications
 - Primary producers worldwide
 - Annual production volumes (tons/year)
 - Feedstock and product market prices (\$/ton)
 - Economic incentives driving process selection
- 2. Typical Process Conditions**
 - Inlet composition (mole fractions or partial pressures)
 - Operating temperature range (K)
 - Operating pressure (bar or atm)
 - Typical reactor conversions (%)
 - Reactor types commonly used
- 3. Competing Feedstock Routes**
 - Alternative ways to make your product
 - Advantages/disadvantages of each route
 - Why your assigned route is preferred
- 4. Reactor Types & Product Distribution**
 - Common reactors for this chemistry (batch, CSTR, PFR, PBR)
 - Why fixed-bed (PBR) is chosen
 - Typical byproduct formation patterns
 - Reaction selectivity observed in industry
- 5. Sustainability Considerations**
 - Environmental impact of current processes
 - Energy efficiency metrics
 - Waste streams and disposal
 - Emerging "green" alternatives

1.2. Reaction Kinetics:

Conduct a **Chemical Abstracts** search for published kinetics of your reaction.

Required:

- At least 2-3 peer-reviewed journal articles with kinetic data
- Rate law expressions (elementary or empirical)
- Kinetic parameters:
 - Rate constant k or pre-exponential factor k_0
 - Activation energy E_a
 - Reaction order with respect to reactants
 - Temperature range of validity

Citation Format: Full AIChE citation required. You can format manually using the provided examples in the project overview document, OR use Zotero (free citation manager with AIChE style)

Calculations

- 2.1. Begin by writing a **balanced** stoichiometric equation for your chemistry. Calculate an overall mass balance for your group's reactor technology. Base your **inlet composition** and **reaction conversion** on typical literature values determined above in section 1.1.2. **State these assumptions!** (Also, in converting your annual production rate to a reactor feed-rate and production rate you should account for scheduled shutdowns of 2 weeks / year).
- 2.2. Using the methodology for flow systems described in Fogler, Chapter 3, section 3.5, develop a **general stoichiometric table** for your reaction system. Give a sample calculation of this mass balance (handwritten).
- 2.3. Develop a Stoichiometric Calculator using Python program
 - INPUT: Stoichiometric coefficients, initial feeds, conversion levels
 - PROCESS: Calculate molar flows using stoichiometry
 - OUTPUT:
 - ✓ Formatted stoichiometric table (like your hand calc)
 - ✓ Molar composition (mole fractions)
 - ✓ Mass composition (mass fractions)
 - ✓ Comparison table (multiple conversions)
 - ✓ Export to CSV for later use
- 2.4. Calculate the heat of reaction (ΔH_{rxn} 298K) to produce your product from all competing feedstocks and **comment** on the differences between feedstocks.
- 2.5. Calculate an overall energy balance using inlet and outlet enthalpies for a generic reactor assuming **typical** reaction conversions determined in section 1.1.2.
 - First, calculate an overall isothermal energy balance and determine heat duty required to operate at a typical temperature (as determined in part 1.1.2).
 - **Second, calculate an overall adiabatic energy balance and determine the outlet temperature.**
 - Use typical inlet temperatures given in either the Kirk-Othmer or McKetta references cited in the overview of the project.
 - Perform calculations using physical property data obtained from references such as the NIST database (NIST WebBook <http://webbook.nist.gov/>). Remember to use heat capacities as a function of temperature.
 - Give a sample calculation of heat capacities, enthalpies and the energy balance and show units in this calculation.
 - Remember to cite references for your thermochemical property data.
1. **UNITS: Please use SI units for all calculations [kg-s-kgmol-m] in this and subsequent memos, as well as in your reactor models.**
2. **Group Duties: Provide a tentative listing of duties for each group member for this memo.**
3. **Hand Calculations: Give hand calculations of each representative calculation in an appendix to this short report. You are not required to type these hand calculations.**

GRADING RUBRIC

Memo 1: 100 Points Total

Section 1: Literature Review (40 pts)

Criterion	Excellent (40)	Good (35)	Acceptable (28)	Poor
Process Background	Comprehensive description of feedstock, product, producers, prices, conditions, competing routes	Good coverage, minor gaps	Covers main points, some missing details	Incomplete, significant gaps
Reactor Types & Technology	Clear explanation of reactor selection, typical conversions, byproduct patterns	Good discussion, adequate justification	Basic description, limited analysis	Vague, incomplete
Sustainability	Thoughtful analysis of environmental impact, energy efficiency, green alternatives	Good discussion of environmental considerations	Mentions sustainability briefly	Not addressed
Kinetics Research	2-3 peer-reviewed sources with clear kinetic parameters, rate laws documented	2 sources, most parameters found	1-2 sources, incomplete kinetic data	Poor quality sources, minimal data
Citations	Complete, properly formatted references (AIChE)	Complete citations, mostly correct format	Most citations present, format issues	Missing citations, poor format

Section 2.1: Hand Calculations (30 pts)

Criterion	Excellent (30)	Good (26)	Acceptable (21)	Poor
Stoichiometric Equation	Balanced correctly, all species identified, units clear	Balanced, all species present, minor issues	Mostly balanced, one or two errors	Unbalanced, missing species
Mass Balance	Complete overall balance, all streams accounted, closed system verified	Good balance, minor gaps	Basic balance, one stream unclear	Incomplete, unchecked
Stoichiometric Table	Clear format, multiple conversions, all calculations shown	Good table, 3-4 conversions shown	Table present, minimal conversions	Minimal or unclear table
Energy Balance Setup	$\Delta H^{\circ}_{\text{rxn}}$, Energy requirements, and outlet temperature calculated correctly, all species and sources cited	Good calculation, proper method	Calculation attempted, one or two errors	Incomplete or incorrect
Assumptions Documented	Clear statement of all assumptions, justified choices	Assumptions listed, mostly justified	Some assumptions stated	Assumptions unclear

Section 2.2: Python Exercise (20 pts)

Criterion	Excellent (20)	Good (17)	Acceptable (14)	Poor
Code Functionality	Runs without errors, produces correct output, handles all conversions	Runs successfully, minor output issues	Runs with one or two bugs, mostly correct	Doesn't run or major errors
Code Quality	Well-commented, functions organized, clear variable names, professional structure	Good comments, logical organization	Minimal comments, works but messy	Hard to understand, confusing
Accuracy	Results match hand calculations exactly, stoichiometry correct	Results match ($\pm 1\%$), methods sound	Some discrepancies, logic mostly correct	Significant errors in calculation
Output & Visualization	Clear formatted tables, easy to read, professional presentation	Readable output, well-organized	Output present, somewhat unclear	Poor formatting, hard to read
Documentation	README with clear instructions, assumptions listed, how to modify	Good documentation, mostly clear	Basic documentation present	Minimal or missing documentation

Presentation & Writing (10 pts)

Criterion	Excellent (10)	Good (9)	Acceptable (7)	Poor
Writing Quality	Clear, professional, no grammar/spelling errors	Generally clear, minor grammar issues	Understandable, some errors	Difficult to read
Organization	Logical flow, easy to follow, good section structure	Well-organized, clear progression	Adequate organization, some sections unclear	Disorganized
Presentation	Professional appearance, proper formatting, visually appealing	Good presentation, readable	Acceptable appearance, could be cleaner	Poor presentation